
THEORETICAL NEUROSCIENCE

TD4: EXCITATORY-INHIBITORY NETWORKS

All TD materials will be made available at https://github.com/yfardella/Theo_Neuro_TD_26_27.

In this tutorial, we will reproduce two results from Murphy & Miller (Neuron, 2009), Balanced Amplification: A New Mechanism of Selective Amplification of Neural Activity Patterns. "In cerebral cortex, ongoing activity absent a stimulus can resemble stimulus-driven activity in size and structure. In particular, spontaneous activity in cat primary visual cortex (V1) has structure significantly correlated with evoked responses to oriented stimuli. This suggests that, from unstructured input, cortical circuits selectively amplify specific activity patterns. Current understanding of selective amplification involves elongation of a neural assembly's lifetime by mutual excitation among its neurons. We introduce a new mechanism for selective amplification without elongation of lifetime: "balanced amplification.""

1 Balanced Amplification

We consider an excitatory-inhibitory network presented in the Fig. 1 and that can be described by the following linear rate model:

$$\tau \frac{d\mathbf{r}}{dt} = -\mathbf{r} + \underline{\underline{W}}\mathbf{r} - \underline{I}. \quad (1)$$

We have

$$\mathbf{r} = \begin{bmatrix} r_E(t) \\ r_I(t) \end{bmatrix}$$

where r_E and r_I are the average firing rates of the E and I populations respectively,

$$\underline{I} = \begin{bmatrix} I_E \\ I_I \end{bmatrix}$$

where I_E and I_I are the external inputs into the E and I populations respectively and

$$\underline{\underline{W}} = \begin{bmatrix} w & -k_I w \\ w & -k_I w \end{bmatrix}$$

where $\underline{\underline{W}}$ is the connectivity matrix. In the following, we assume that inhibition balances excitation, hence, $\underline{\underline{k}}_I > 1$.

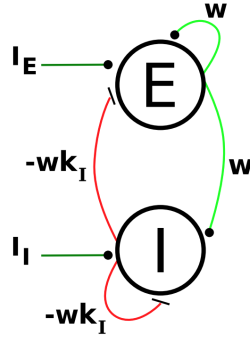


Figure 1

1.1 Network relaxation

In this subsection, we assume that there is no input into the populations, i.e. $\underline{I} = \underline{0}$ and that $r_E(0) = 1$ and $r_I(0) = 0$.

1. If we consider a single excitatory population recurrently exciting itself with strength w , what are the possible steady-state solutions depending on the value of w ?

The ODE governing the dynamics of $r_E(t)$ is

$$\tau \frac{dr_E}{dt} = (w - 1)r_E.$$

If $w < 1$, r_E decreases exponentially towards 0. If $w > 1$, r_E goes to infinity.

2. Find the eigenvalues and eigenvectors of $\underline{W}$. Comment on the sign of the eigenvalues.

We need to find the roots the characteristic polynomial,

$$\det(\lambda \underline{I} - \underline{W}) = \begin{vmatrix} \lambda - w & k_I w \\ -w & \lambda + k_I w \end{vmatrix} = \lambda(\lambda + w(k_I - 1)) = 0.$$

The 2 eigenvalues are $\lambda_0 = 0$ and $\lambda_+ = -w(k_I - 1) = -w_+$, letting $w_+ = w(k_I - 1)$. For the eigenvalue $\lambda_0 = 0$, we compute the eigenvector,

$$\begin{bmatrix} w & -k_I w \\ w & -k_I w \end{bmatrix} \cdot \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} wA - k_I wB \\ wA - k_I wB \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

hence, $A = k_I B$. Therefore, an eigenvector associated to the eigenvalue $\lambda_0 = 0$ is

$$\underline{p}^0 = \begin{bmatrix} k_I \\ 1 \end{bmatrix}.$$

For the eigenvalue $\lambda_+ = -w_+$, we compute the eigenvector,

$$\begin{bmatrix} w & -k_I w \\ w & -k_I w \end{bmatrix} \cdot \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} -w_+ \\ -w_+ \end{bmatrix},$$

hence, $A = k_I B + 1 - k_I$. Therefore, an eigenvector associated to the eigenvalue $\lambda_+ = -w_+$ is

$$\underline{p}^+ = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

These eigenvectors being non-orthogonal, the interpretability of the network dynamics in terms of the eigenvectors can be deceptive. Therefore, we propose to study the dynamics in the following orthogonal basis, letting,

$$\underline{p}^+ = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and

$$\underline{p}^- = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

3. Compute $\underline{W} \cdot \underline{p}^+$ and $\underline{W} \cdot \underline{p}^-$. Comment.

We can compute the dot product directly,

$$\underline{W} \cdot \underline{p}^+ = \begin{bmatrix} -w_+ \\ -w_+ \end{bmatrix},$$

since $\underline{p}^+$ is the eigenvector computed in the previous question and

$$\underline{W} \cdot \underline{p}^- = \begin{bmatrix} w + k_I w \\ w + k_I w \end{bmatrix} = w(1 + k_I) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = w_{FF} \underline{p}^+,$$

letting $w_{FF} = w(1 + k_I)$ (the Feed Forward connection strength). We observe that $\underline{p}^-$ is not an eigenvector of $\underline{W}$.

4. Decompose $\underline{r}(t)$ as a sum of these 2 vectors, $\underline{r}(t) = \underline{p}^+ r_+(t) + \underline{p}^- r_-(t)$

Looking in the r_E and r_I basis:

$$\underline{r} = \begin{bmatrix} r_E \\ r_I \end{bmatrix} = \begin{bmatrix} r_+(t) + r_-(t) \\ r_+(t) - r_-(t) \end{bmatrix},$$

hence,

$$2r_+(t) = r_E + r_I \Rightarrow r_+(t) = \frac{1}{2}(r_E + r_I)$$

and

$$2r_-(t) = r_E - r_I \Rightarrow r_-(t) = \frac{1}{2}(r_E - r_I).$$

5. Compute the time course of $r_+(t)$ and $r_-(t)$.

We need to derive the ODEs governing the evolution of $r_+(t)$ and $r_-(t)$ using the decomposition $\underline{r}(t) = \underline{p}^+ r_+(t) + \underline{p}^- r_-(t)$ in Eq. 1:

$$\begin{aligned} \tau \frac{dr}{dt} &= \tau \frac{dr_+}{dt} \underline{p}^+ + \tau \frac{dr_-}{dt} \underline{p}^- \\ &= -r_+ \underline{p}^+ - r_- \underline{p}^- + r_+ \underline{W} \cdot \underline{p}^+ - r_- \underline{W} \cdot \underline{p}^-, \end{aligned}$$

therefore,

$$\tau \frac{dr}{dt} = (-r_+(1 + w_+) + w_{FF} r_-) \underline{p}^+ - r_- \underline{p}^-. \quad (2)$$

Knowing that the eigenvectors are orthogonal:

$$\tau \frac{dr_+}{dt} = -r_+(1 + w_+) + w_{FF}r_- \quad (3)$$

and

$$\tau \frac{dr_-}{dt} = -r_- \quad (4)$$

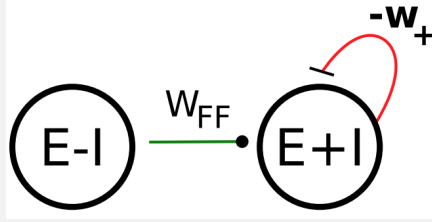


Figure 2: Reduced network.

The solution for $r_-(t)$ is straightforward:

$$r_-(t) = \frac{1}{2} \exp\left(-\frac{t}{\tau}\right),$$

using $r_-(0) = 1/2$. To compute $r_+(t)$, we use the Ansatz,

$$r_+(t) = C(t) \exp\left(-(1 + w_+)\frac{t}{\tau}\right),$$

with $C(0) = 1/2$. Then, in Eq. 3:

$$\begin{aligned} \tau \frac{dC}{dt}(t) \exp\left(-(1 + w_+)\frac{t}{\tau}\right) &= w_{FF}r_- \\ \Rightarrow \frac{dC}{dt}(t) &= \frac{w_{FF}}{2\tau} \exp\left(w_+\frac{t}{\tau}\right). \end{aligned}$$

Integrating over t , we get,

$$\begin{aligned} C(t) - C(0) &= C(t) - \frac{1}{2} = \int_0^t \frac{w_{FF}}{2\tau} \exp\left(w_+\frac{s}{\tau}\right) ds \\ &= \frac{w_{FF}}{2w_+} \left(\exp\left(w_+\frac{t}{\tau}\right) - 1 \right), \end{aligned}$$

Therefore,

$$r_+(t) = \frac{1}{2} \left(\frac{w_{FF}}{w_+} \exp\left(-\frac{t}{\tau}\right) - \left(\frac{w_{FF}}{w_+} - 1 \right) \exp\left(-(1 + w_+)\frac{t}{\tau}\right) \right).$$

6. Setting $k_I = 1.1$, draw the time course of $r_E(t)$ for $w = 3/4$ and $w = 4$.

The time course of $r_E(t)$ is given by

$$r_E(t) = r_+ + r_- = \frac{1}{2} \left(\left(\frac{w_{FF}}{w_+} + 1 \right) \exp \left(-\frac{t}{\tau} \right) - \left(\frac{w_{FF}}{w_+} - 1 \right) \exp \left(-(1 + w_+) \frac{t}{\tau} \right) \right).$$

We then compute the root t_r of the time derivative of $r_E(t)$:

$$r'_E(t_r) = -\frac{1}{2\tau} \left(\left(\frac{w_{FF}}{w_+} + 1 \right) \exp \left(-\frac{t_r}{\tau} \right) - (1 + w_+) \left(\frac{w_{FF}}{w_+} - 1 \right) \exp \left(-(1 + w_+) \frac{t_r}{\tau} \right) \right),$$

which is given by

$$\begin{aligned} (w_{FF} + w_+) &= (1 + w_+) (w_{FF} - w_+) \exp \left(-w_+ \frac{t_r}{\tau} \right) \\ \Rightarrow t_r &= \frac{\tau}{w_+} \log \left(\frac{(1 + w_+)(w_{FF} - w_+)}{(w_{FF} + w_+)} \right). \end{aligned}$$

We observe that t_r and $r'_E(0)$ have the same sign.

With the given parameters, when $w = 4$, $t \simeq 0.6$ and $r'_E(0) > 0$.

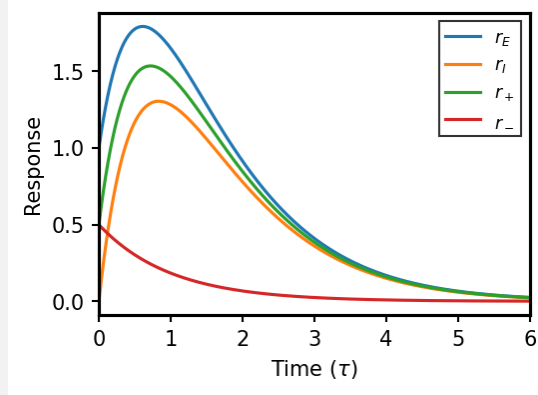


Figure 3: Response of r_E and r_I for $w = 4$

When $w = 3/4$, $t < 0$ and $r'_E(0) < 0$.

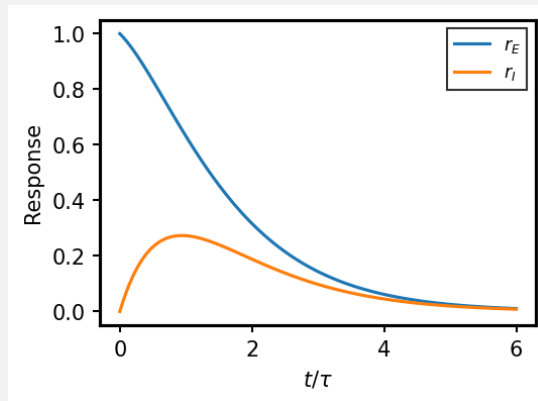


Figure 4: Response of r_E and r_I for $w = 3/4$

7. What does balanced amplification mean?

Balanced amplification arises when feedback inhibition stabilizes strong recurrent excitation. In particular when the recurrent excitation without balance would lead to diverging excitatory activity.

1.2 Response to a pulse input

In this subsection, we assume that $r_E(0) = 0$, $r_I(0) = 0$ and the input to the network is a pulse at time 0:

$$\underline{I}(t) = \begin{bmatrix} \tau\delta(t) \\ 0 \end{bmatrix}.$$

8. How are the ODEs governing the time course of $r_+(t)$ and $r_-(t)$ obtained Q5 modified?

In Q5, we had $\underline{I} = 0$. Noting that $\underline{I} = \tau\delta(t)(\underline{p}^+ + \underline{p}^-)/2$, we add this term in Eq. 2:

$$\tau \frac{dr}{dt} = (-r_+(1 + w_+) + w_{FF}r_-) \underline{p}^+ - r_- \underline{p}^- + \frac{\tau\delta(t)}{2}(\underline{p}^+ + \underline{p}^-).$$

Then,

$$\tau \frac{dr_+}{dt} = -r_+(1 + w_+) + w_{FF}r_- + \frac{\tau\delta(t)}{2} \quad (5)$$

and

$$\tau \frac{dr_-}{dt} = -r_- + \frac{\tau\delta(t)}{2}. \quad (6)$$

9. Compute the time course of $r_+(t)$ and $r_-(t)$. Comment.

Making the Ansatz

$$r_-(t) = C(t) \exp\left(-\frac{t}{\tau}\right),$$

with $C(0) = 0$. We have, in Eq. 6,

$$\begin{aligned} \tau \frac{dC}{dt}(t) \exp\left(-\frac{t}{\tau}\right) &= \frac{\tau\delta(t)}{2} \\ \Rightarrow \frac{dC}{dt}(t) &= \frac{\delta(t)}{2} \exp\left(\frac{t}{\tau}\right). \end{aligned}$$

Then,

$$C(t) = \int_0^t \frac{\delta(s)}{2} \exp\left(\frac{s}{\tau}\right) ds = \frac{1}{2} \exp\left(\frac{0}{\tau}\right) = \frac{1}{2}.$$

Therefore,

$$r_-(t) = \frac{1}{2} \exp\left(-\frac{t}{\tau}\right).$$

We can compute $r_+(t)$ using the same method as in Q5:

$$\frac{dC}{dt}(t) = \frac{w_{FF}}{2\tau} \exp\left(w_+ \frac{t}{\tau}\right) + \frac{\delta(t)}{2} \exp\left((1 + w_+) \frac{t}{\tau}\right).$$

Then,

$$r_+(t) = \frac{1}{2} \left(\frac{w_{FF}}{w_+} \exp \left(-\frac{t}{\tau} \right) - \left(\frac{w_{FF}}{w_+} - 1 \right) \exp \left(-(1 + w_+) \frac{t}{\tau} \right) \right).$$

The solutions are the same as in Q5, hence, the dynamic of the network in response to a delta impulse into the excitatory population at time 0 is similar to the dynamic of the network with no input and $r_E(0) = 1$ and $r_I(0) = 0$.